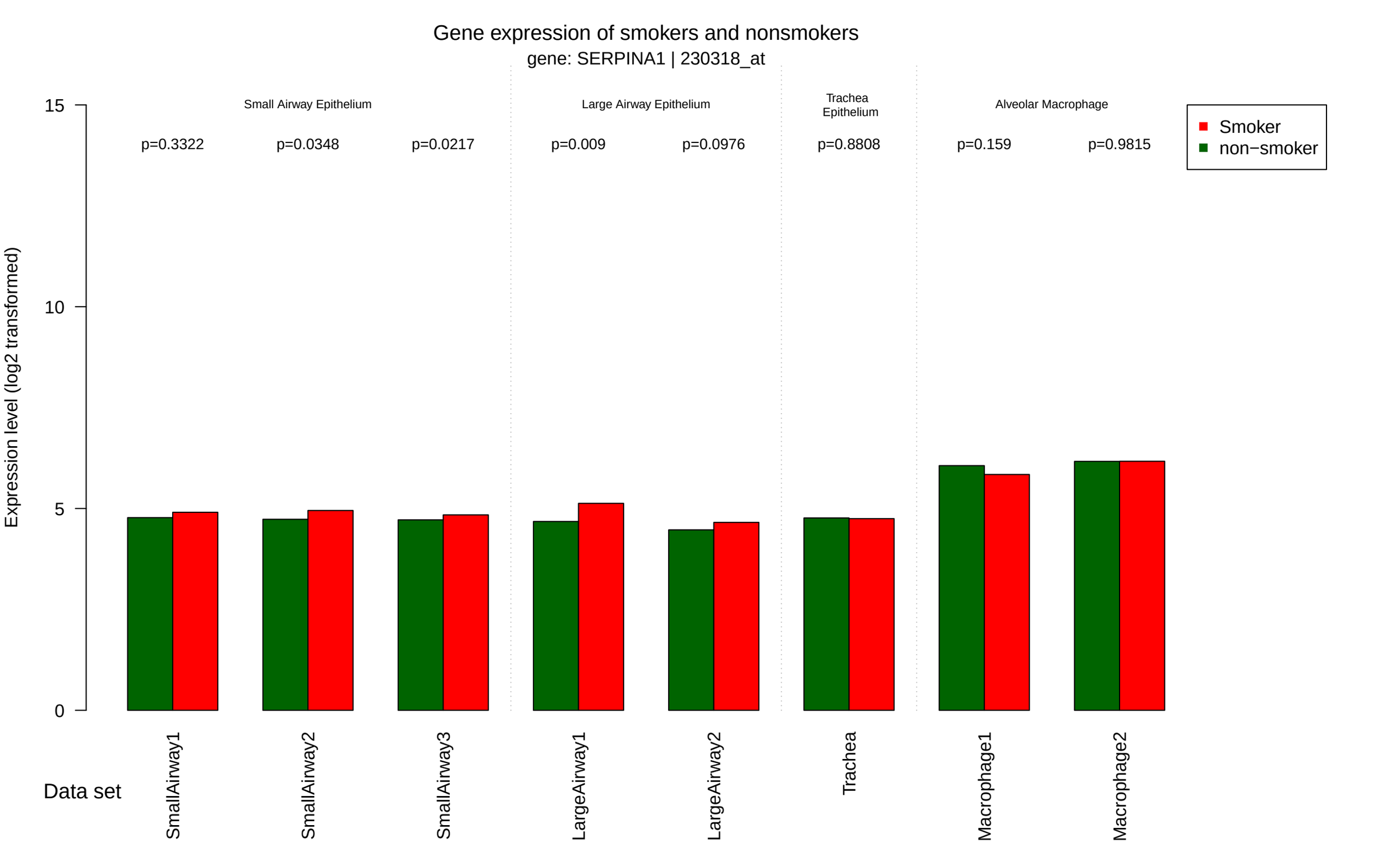
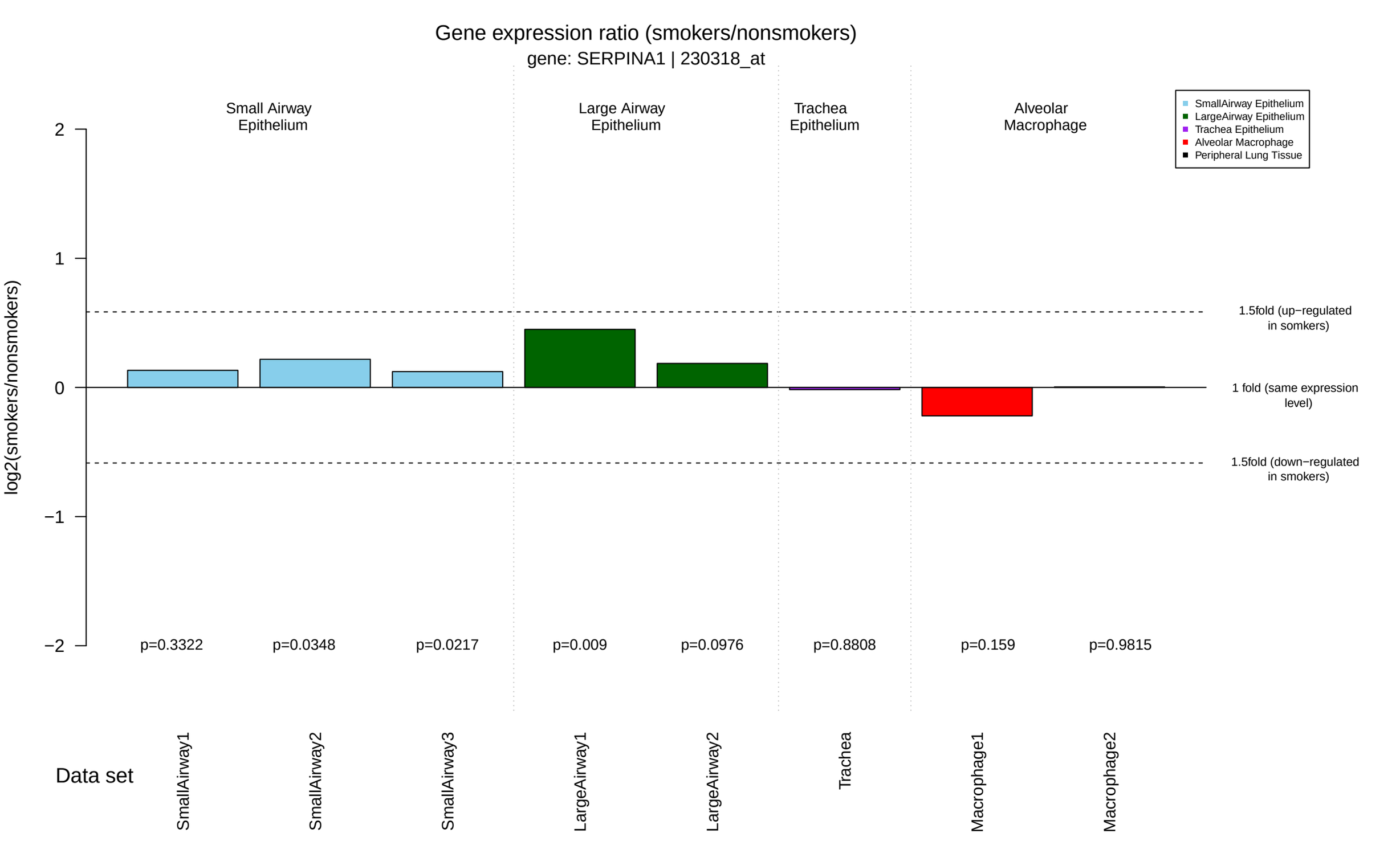
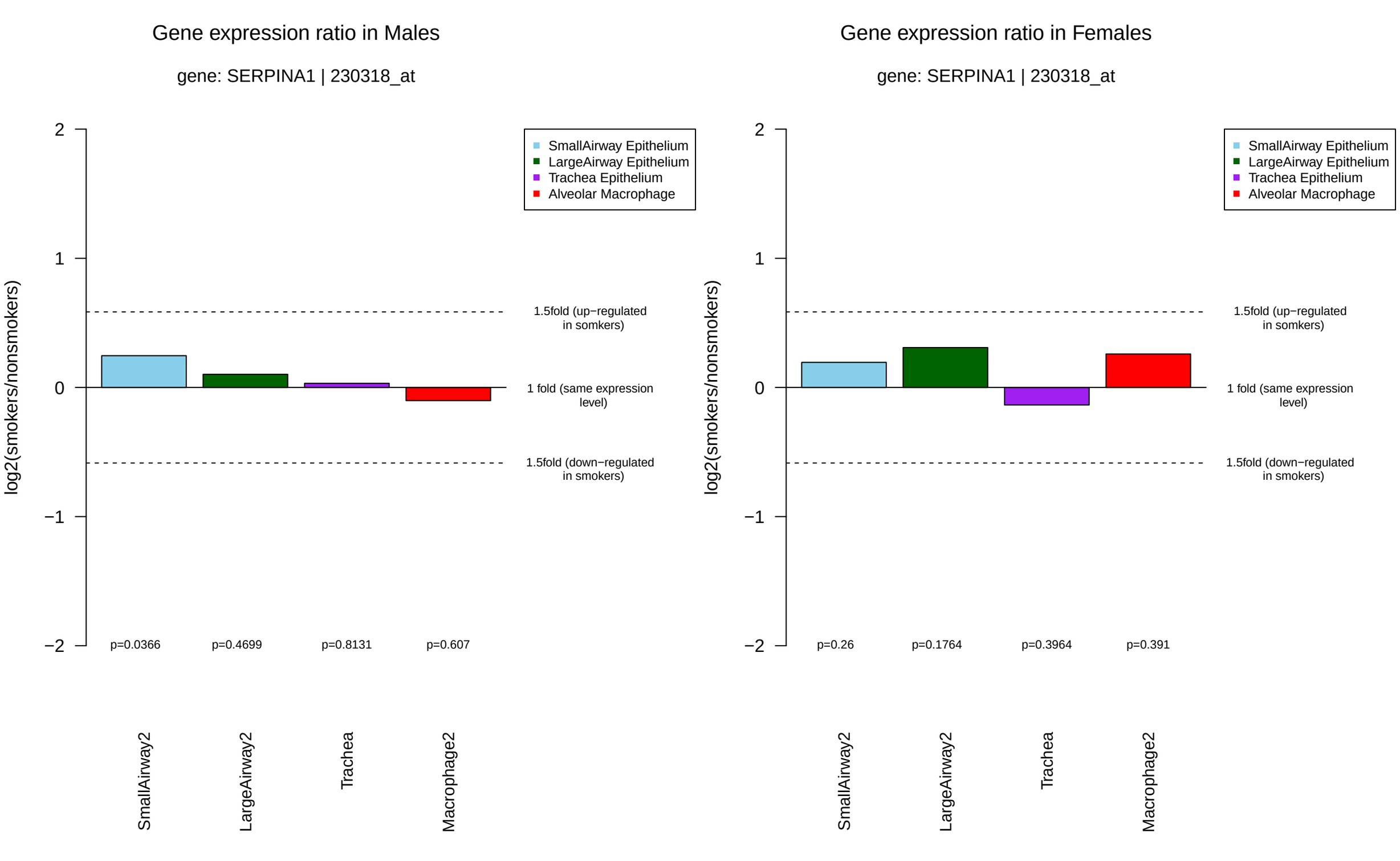
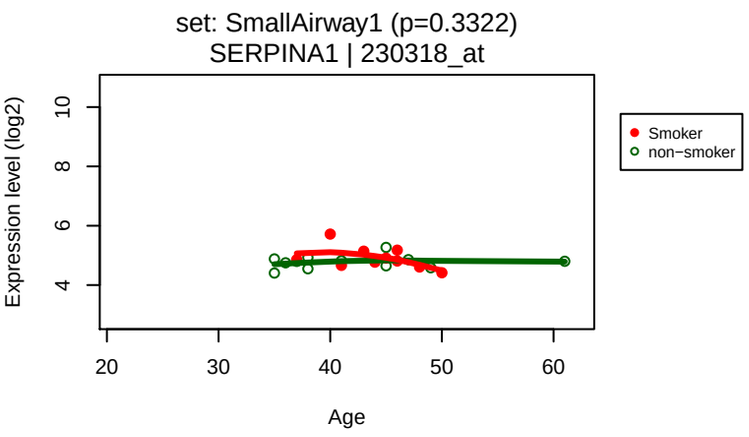
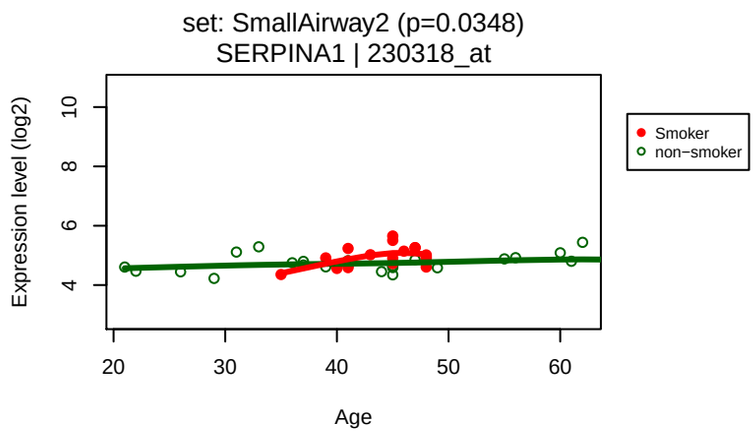
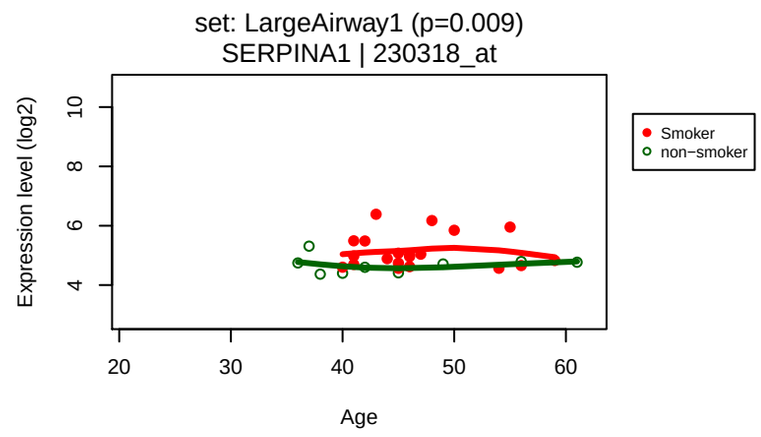

Gene expression of smokers and nonsmokers
gene: SERPINA1 | 230318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3322
p=0.0348
p=0.0217
p=0.009
p=0.0976
p=0.8808
p=0.159
p=0.9815
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

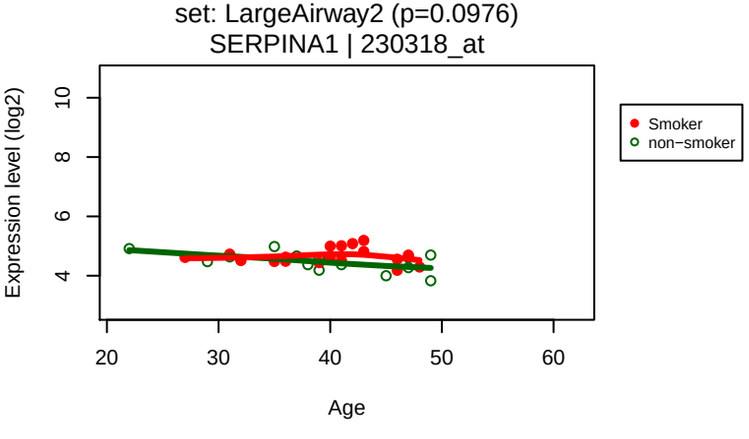

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA1 | 230318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3322
p=0.0348
p=0.0217
p=0.009
p=0.0976
p=0.8808
p=0.159
p=0.9815
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

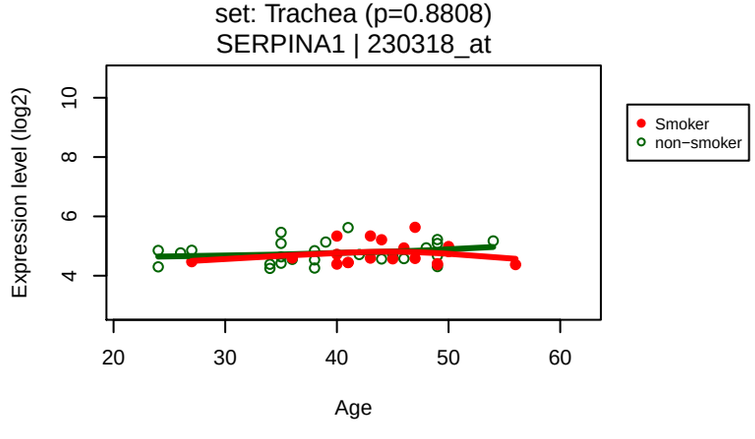

Gene expression ratio in Males
gene: SERPINA1 | 230318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0366
p=0.4699
p=0.8131
p=0.607
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINA1 | 230318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.26
p=0.1764
p=0.3964
p=0.391
SmallAirway2
LargeAirway2
Trachea
Macrophage2

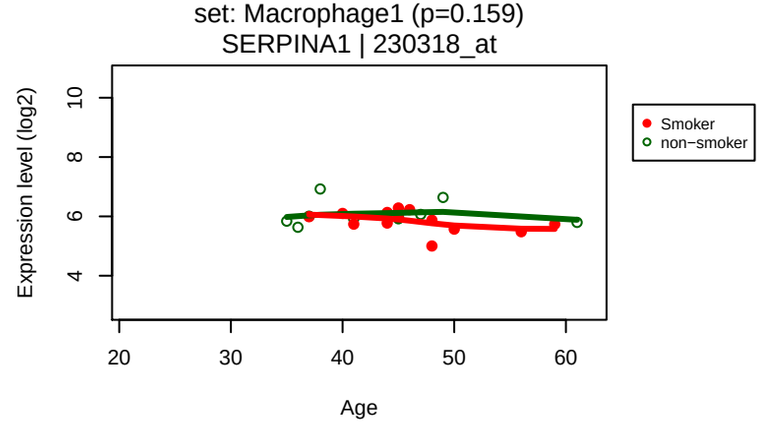

set: SmallAirway1 (p=0.3322)
SERPINA1 | 230318_at
Expression level (log2)
Age
Smoker
non−smoker

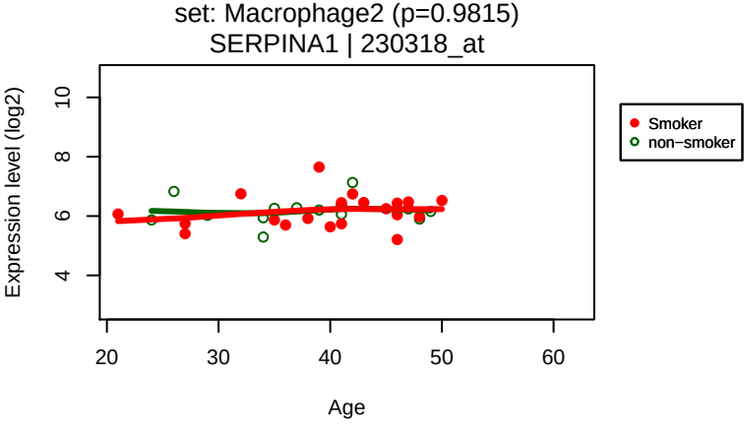

set: SmallAirway2 (p=0.0348)
SERPINA1 | 230318_at
Expression level (log2)
Age
Smoker
non−smoker

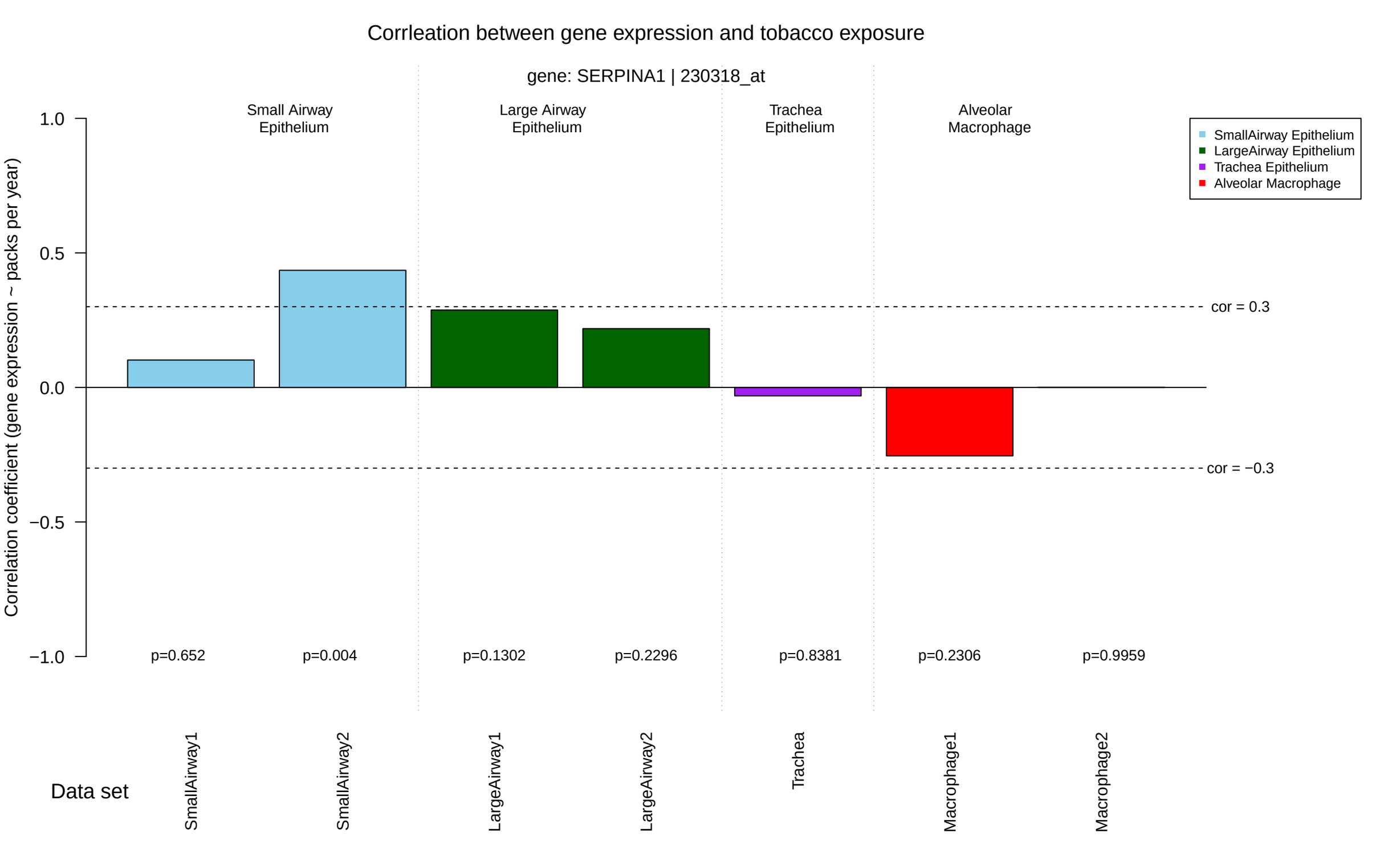

set: LargeAirway1 (p=0.009)
SERPINA1 | 230318_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0976)
SERPINA1 | 230318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8808)
SERPINA1 | 230318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.159)
SERPINA1 | 230318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9815)
SERPINA1 | 230318_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA1 | 230318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.652
p=0.004
p=0.1302
p=0.2296
p=0.8381
p=0.2306
p=0.9959
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2